Power Budget

Team Number: **205**
Project Name: **BAMF**
Team Member Name: **Jacob Silber**

All Major Components	Component Name	Part Number	Supply Voltage Range	Qty	Absolute Maximum Current	Total Current	Unit
	Photoresistor	PDV-P8103	0-150V	1	2.37	2.37	mA
	Op Amp	MCP6004-I/P	1.8-6V	1	92	92	mA
	Curiosity Nano	PIC18F57Q43	1.8-5.5V	1	500	500	mA

5V Power Rail	Component Name	Part Number	Supply Voltage Range	Qty	Absolute Maximum Current	Total Current	Unit
	Photoresistor	PDV-P8103	0-150V	1	2.37	2.37	mA
	Op Amp	MCP6004-I/P	1.8-6V	1	92	92	mA
	Curiosity Nano	PIC18F57Q43	1.8-5.5V	1	500	500	mA
					Subtotal	594.37	mA
					Safety Margin	25%	
					Total Current Required on 5V rail	742.9625	mA
Regulator	5V Regulator	L7805ABV-DG	0-35V	1	1500	1500	mA
			Total Remaining Current on 5V rail			757.0375	mA

External Power Source 1	Component Name	Part Number	Supply Voltage Range	Output Voltage	Absolute Current Maximum	Total Current	Unit
Power Source 1 Selection	Plug-in Wall Supply	NGE30U12-P1J	85-265VAC	12	2500	2500	mA
Power rails connected to External Power Source	5V Regulator	L7805ABV-DG	0-35V	5V	1500	1500	mA
		Total Remaining Current Available on External Power Source 1				1000	mA